

Synthetic Market Phase 8

Contextual relation follow-through on the Phase 7 relation-invariance result. This phase keeps the anchor pair numerically fixed and asks whether the pair’s relation survives harder contextual roster pressure.

DATE	PROMPTS	DESIGN	OBJECT
21 March 2026	384 market-only	4 scenarios × 2 styles × 4 layouts × 4 rosters × 3 scales	contextual anchor-pair relation

MAIN READ

Phase 8 breaks most of the easy Phase 7 relation win: primitive factors stay explicit, but contextual relation identity is only weakly preserved.

This is a useful negative result. Once the anchor pair is held numerically fixed and the surrounding roster changes what that pair means, relation retrieval mostly collapses. The one family that still shows a modest surviving signal is `paired_cluster_context`.

BEST PRIMITIVE R ²	BEST RELATION	BEST RANK CTRL	BEST SCALE CTRL
0.99993 row_mean L1	0.03273 paired cluster, style-only	0.00459 paired cluster	0.01397 paired cluster

Why Phase 8 Exists

Phase 7 showed that relation identity beat rank-bucket and scale-bucket confounds. That was encouraging, but still likely too easy.

Phase 8 tightens the question:

- keep the anchor pair numerically fixed
- move it through harder contextual rosters
- keep the same nuisance axes from Phase 7

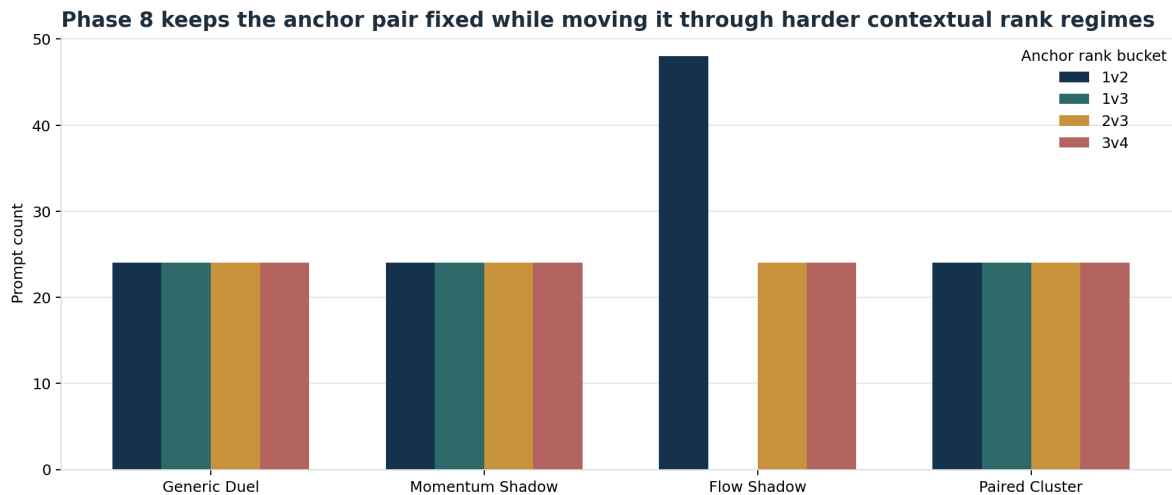
The new target is not “can the model preserve a simple pair?” but “does the meaning of that pair stay stable when the surrounding market changes?”

Primitive Factors Still Dominate The Rows

Variable	Best State	Metric	Layer
pct_5m	row_mean	R ² 0.99992	L1
net_flow_5m	row_mean	R ² 0.99991	L1
unique_traders_5m	row_mean	R ² 0.99991	L1
top20_holder_pct	row_mean	R ² 0.99990	L1
attractiveness_score	row_mean	R ² 0.99993	L1
risk_adjusted_score	row_mean	R ² 0.99993	L1

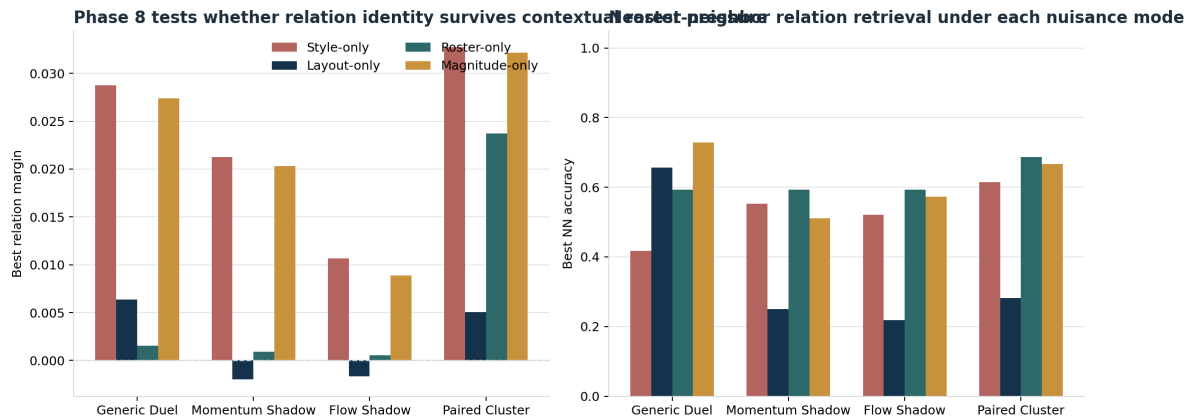
Primitive market factors remain almost perfectly linearly explicit. So Phase 8 is not showing a loss of market information; it is showing a failure of the **relation object** under harder contextual pressure.

The Contextual Rank Regimes Are Real



The anchor pair is numerically fixed, but the surrounding roster moves it across different rank regimes. This is a contextual market test, not a simple reformatting exercise.

Relation Identity Mostly Collapses



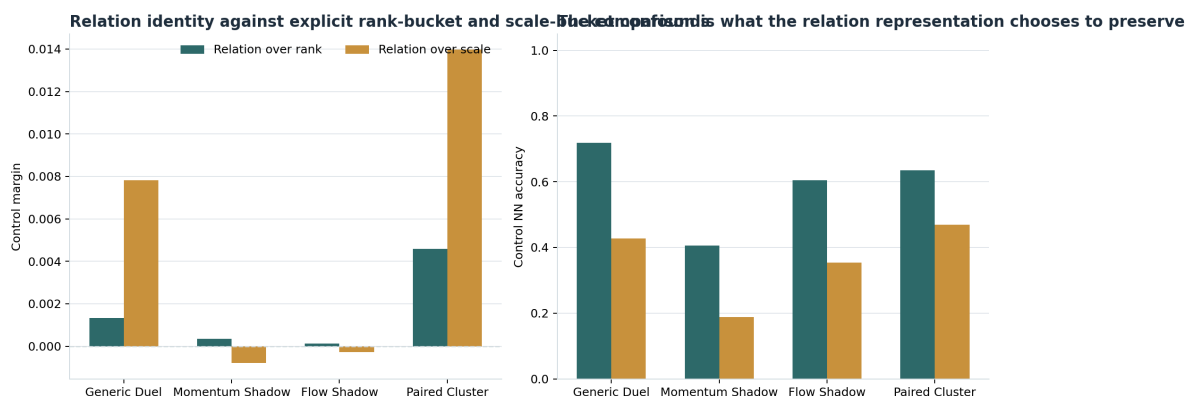
Phase 8 margins are much smaller than Phase 7. The remaining signal is modest and scenario-specific rather than broad and invariant.

Scenario	Best Mode	Best Margin	NN Acc.
generic duel	style-only	0.02874	0.4167
momentum shadow	style-only	0.02126	0.5521
flow shadow	style-only	0.01065	0.5208
paired cluster	style-only	0.03273	0.6146

The large Phase 7 margins are gone. What remains is weak and uneven:

- paired_cluster_context is the only scenario that still looks somewhat coherent
- generic_duel_context and momentum_shadow_context retain only modest signal
- flow_shadow_context is nearly gone

Hard Controls Expose The Failure Mode



The hard question is whether relation identity beats explicit contextual confounds. In Phase 8, those margins are weak almost everywhere.

Scenario	Rank Margin	Scale Margin	Best State
generic duel	0.00134	0.00782	row_mean L1 / row_eos L36
momentum shadow	0.00034	-0.00080	row_eos L4
flow shadow	0.00013	-0.00027	row_mean L1 / row_eos L36
paired cluster	0.00459	0.01397	row_eos L35 / L36

The failure is most obvious in the harder axes:

- layout_only is weak or negative in three of four scenarios
- roster_only is near zero except for paired_cluster_context
- rank_ctrl is almost zero everywhere
- scale_ctrl is only meaningfully positive for paired_cluster_context

This is the real Phase 8 result: the broader contextual meaning of a fixed pair is not a broadly stable invariant representation.

Interpretation

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The model still carries primitive market factors very clearly and can still solve direct pairwise comparisons with ease.

LIMITATION

A fixed anchor-pair relation is not a broadly robust market representation object once contextual roster pressure is introduced. Phase 7 overstated the strength of that object.

The strongest honest reading is:

- primitive local factors are real
- direct pairwise preference is real
- but contextual relation identity is not generally stable

This is useful because it tells us what **not** to overclaim.

What To Do Next

Phase 8 suggests the next object should not be “the identity of a fixed pair.”

Better next directions:

1. set-level market geometry
2. factor-difference representations like delta momentum, delta participation, and delta concentration
3. context-conditioned relation families rather than one frozen anchor pair

4. explicit follow-up on `paired_cluster_context`, the only scenario that still shows a meaningful surviving signal

Conclusion

Phase 8 is a negative result in the right place.

It keeps the good part of the relation-first program:

- focus on upstream comparative structure, not end actions

But it shows the limit of the current object:

- fixed anchor-pair relation is too brittle under real contextual roster pressure

That is progress. It means the next market-representation experiment should move from pair identity to richer context-sensitive relational structure.